

Al Hamadani, Amir Hussein R.

alhamadani.amirrara@gmail.com | +63 910-536-6224 | github.com/alhamadaniamir

SUMMARY

Computer Engineering Student focused on solving practical problems, reducing manual work, and improving operational efficiency. Experienced in building IoT systems, robotic controls, and full-stack applications. Combines rapid learning with in-depth research, systematic debugging, and careful testing to develop reliable solutions. Brings high standards for correctness, security, and maintainability. Driven by a need to understand how things actually work under the hood and build technology that addresses real user and operational needs.

TECHNICAL SKILLS

Languages: C/C++, Python, Javascript, Rust, MATLAB, MicroPython, Java, Dart, PHP, Bash/Shell, Verilog.

Frameworks: React, Node.js, Flask, FastAPI, Django, React Native, TensorFlow, Pytorch, Flutter, Tailwind CSS.

Database: MongoDB, MySQL, SQLite.

Embedded & IoT: Raspberry Pi, ESP32, ESP32 S3, ESP32-CAM, Arduino nano, Arduino Uno, Vercel, JSON, Firebase Realtime Database, Blynk.

CyberSecurity: SIEM, Wireshark, JWT Authentication, tcpdump.

Robotics & Automation: Drake, Forward & Inverse Kinematics, Motion Planning, Robot Dynamics, Control Systems.

AI & Tools: Google Gemini API, Computer Vision, Image Classification, Scikit-learn, Arch Linux, Git.

KEY PROJECTS

Computer Vision-Based Fruit Detection Automated IoT Weighing Scale with AI-Driven Sales Forecasting and Data Analytics | *ESP32, ESP32-CAM, Google Gemini, Multimodal*

- Integrated ESP32-CAM with Gemini through a Rust API to identify 23 fruit labels, including three mango varieties.
- Implemented automatic pricing, five-point load-cell calibration, and ESP-NOW communication with Firebase sales synchronization and upload retries.

FruityVens, IoT Scale Companion App | *Flutter, Dart, HTTP/JSON*

- Built a mobile interface to monitor detected fruit, weight, and price, and confirm sales through ESP32 REST endpoints.
- Added configurable device connectivity, automatic status updates every three seconds, and retrieval of the latest sale record.
- Implemented bearer-token authentication for the fruit-identification API, kept Gemini credentials server-side, and redacted API keys from error logs.

Robotic Arm Motion Control System | *Arduino, C++, Servo, AccelStepper*

- Developed firmware to sequence six servos and a stepper motor, controlling base, shoulder, elbow, wrist, and gripper movements.
- Implemented mirrored base-servo motion, configurable joint travel limits, gradual servo positioning, and acceleration-controlled stepper movement.

Ballclub, Basketball Session Management Platform | *Vue 3, FastAPI, MongoDB, Tailwind CSS, Chart.js*

- Built session management with shareable join links, owner approvals, numbered player queues, and configurable capacity limits.
- Added payment tracking, member attendance histories, revenue dashboards, and CSV exports for sessions and member records.
- Implemented JWT authentication, bcrypt password hashing, and request rate limits, with 56 automated API tests covering authentication and club workflows.

EDUCATION

University of Science and Technology of Southern Philippines

2022-2024

Bachelor of Science in Computer Engineering

Relevant Coursework: Data Structures & Algorithms Analysis, Object Oriented Programming, Electrical Circuits, Operating Systems, Software Development, Project Management, Numerical Methods and Analysis

Phinma Cagayan de Oro College

2024-2027

Bachelor of Science in Computer Engineering

Relevant Coursework: Embedded Systems, Microprocessors, Computer Networks and Security, Logic Circuits and design, Feedback and Control Systems, Digital Signal Processing

Capitol University

2018-2020

Senior High School strand STEM-IT